

MATH 458 Notes : Honours Differential Geometry

Samy Lahlou

These notes are based on lectures given by Professor Jean Pierre Mutanguha at McGill University during the Winter 2026 semester. These lectures will introduce the local & global theory of curves and surfaces in two and three dimensions. As a disclaimer, it is more than possible that I made some mistakes. Feel free to correct me or ask me anything about the content of this document at the following address : samy.lahloukamal@mail.mcgill.ca

Contents

1	Preliminaries	1
1.1	Linear Algebra	1
1.2	Topology	3
1.3	Vector Calculus	4
2	Paths & Curves	7
2.1	Paths	7
2.2	Reparametrizations of paths	9
2.3	Curves	11
2.4	Curvature	12
2.5	Plane Paths	15
2.6	Space paths	18
3	Surfaces	21
3.1	Definitions and Basic Properties	21
3.2	Orientation	23
3.3	Functions between Surfaces	25
3.4	The First Fundamental Form	27
3.5	Gaussian and Mean Curvature	29
3.6	The Second Fundamental Form	30

1 Preliminaries

1.1 Linear Algebra

In this course, we will focus on the vector space

$$\mathbb{R}^n = \{(x_1, \dots, x_n) : x_i \in \mathbb{R}\},$$

especially $\mathbb{R}^2$ and $\mathbb{R}^3$ even if everything that will be covered can be generalized for $\mathbb{R}^n$ where $n \geq 2$ is arbitrary. We will also care about the standard dot product on $\mathbb{R}^n$, defined by

$$\mathbf{v} \cdot \mathbf{w} = \sum_i v_i w_i.$$

This is an example of an *inner product*, i.e., a symmetric function $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, bilinear, and positive definite. Notice that for any invertible linear transformation T from $\mathbb{R}^n$ to $\mathbb{R}^n$, we can define a new inner product $b_T : (v, w) \mapsto Tv \cdot Tw$. It turns out that every inner product on $\mathbb{R}^n$ is of this form for some linear transformation T . Using the inner product, we can define the usual notions of Euclidean norm and Euclidean distance in the vector space $\mathbb{R}^n$:

$$\|v\| = \sqrt{v \cdot v}$$

and

$$d(v, w) = \|v - w\|.$$

The goal now is to define what it means for a transformation to be a rigid motion. Intuitively, a rigid motion is like a translation or a rotation, it preserves distances. This motivates the following definition.

Definition. A linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is *orthogonal* if the following conditions are satisfied:

- $T(v) \cdot T(w) = v \cdot w$ for all $v, w \in \mathbb{R}^n$,
- $\|T(v)\| = \|v\|$ for all $v \in \mathbb{R}^n$,
- $d(T(v), T(w)) = d(v, w)$ for all $v, w \in \mathbb{R}^n$.

It turns out that the three conditions in this definition are equivalent. This should not be surprising since the notions of norm and distance are both defined in terms of the dot product. Let's prove this.

Proposition 1.1.1. The following are equivalent:

- (1) $T(v) \cdot T(w) = v \cdot w$ for all $v, w \in \mathbb{R}^n$,
- (2) $\|T(v)\| = \|v\|$ for all $v \in \mathbb{R}^n$,
- (3) $d(T(v), T(w)) = d(v, w)$ for all $v, w \in \mathbb{R}^n$.

Proof. Condition (1) implies condition (2) because if T preserves the dot product, then $T(v) \cdot T(v) = v \cdot v$ for all $v \in V$. Taking the square root on both sides gives us precisely condition (2). Condition (2) implies condition (3) because replacing v with $v - w$ in condition (2) gives us $\|T(v - w)\| = \|v - w\|$ which is equivalent to $d(T(v), T(w)) = d(v, w)$. Similarly, condition (3) implies condition (2) by taking $w = 0$.

Finally, we need to show that conditions (2) and (3) imply condition (1). To do so, let $v, w \in V$, then $d(T(v), T(w)) = d(v, w)$. Squaring both sides and using the definition of the distance gives us

$$(v - w) \cdot (v - w) = (T(v) - T(w)) \cdot (T(v) - T(w)).$$

By symmetry and bilinearity of the dot product, we can expand both sides to get

$$v \cdot v - 2(v \cdot w) + w \cdot w = T(v) \cdot T(v) - 2(T(v) \cdot T(w)) + T(w) \cdot T(w).$$

By condition (2), $v \cdot v = T(v) \cdot T(v)$ and $w \cdot w = T(w) \cdot T(w)$, hence, after cancelling the terms:

$$v \cdot w = T(v) \cdot T(w)$$

which is exactly condition (1). ■

For the moment, an orthogonal transformation is not exactly what we have in mind when we talk about rigid motions. For example, the transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (-x, y)$ is an orthogonal transformation, but it flips the plane so it is not a rigid motion. The following proposition makes the phenomenon more precise.

Proposition 1.1.2. If T is an orthogonal transformation, then $\det(T) = \pm 1$.

Proof. Let $e_1, \dots, e_n$ be the standard basis in $\mathbb{R}^n$. Let $A = (a_{ij})_{ij}$ be the matrix representation of the orthogonal transformation T , then $Te_i = \sum_k a_{ik} e_k$. Since T is orthogonal, then it preserves the inner product. It follows that $Te_i \cdot Te_j = \delta_{ij}$. But on the other hand, we have that $Te_i \cdot Te_j = \sum_k a_{ik} a_{jk}$. Next, consider the matrix AA^T , its coefficient with index ij is equal to $\sum_k a_{ik} b_{kj}$ where $A^T = (b_{ij})_{ij}$. But since $b_{ij} = a_{ji}$ holds for all i and j , then the coefficient with index ij in AA^T is

$$\sum_k a_{ik} a_{jk} = Te_i \cdot Te_j = \delta_{ij}.$$

In other words, $AA^T = I_n$. Taking the determinant on both sides gives us $\det(AA^T) = 1$. Since the determinant preserves the product of matrices, and $\det(A^T) = \det(A)$, then the last equation is equivalent to $\det(A)^2 = 1$. Therefore, $\det(T) = \det(A) = \pm 1$. ■

We can think of a transformation with negative determinant in $\mathbb{R}^2$ as a transformation that flips the plane. More generally, such a transformation can be thought as changing the orientation of the space. This motivates the next definition.

Definition. A linear transformation T is *orientation-preserving* if $\det(T) > 0$.

An orthogonal orientation-preserving transformation can be thought as rotations about the origin. Hence, if we combine the last two definitions, it would be tempting to say that we would get the definition of a rigid motion. However, our definitions only apply to linear transformations, and not translations for example. This is precisely what is the left to add in our definition of a rigid motion.

Definition. A function $M : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a *rigid motion* if there is an element $a \in \mathbb{R}^n$ and an orthogonal orientation-preserving transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $M(v) = T(v) + a$ for all $v \in \mathbb{R}^n$. Equivalently, a rigid motion is an affine linear map where linear part is orthogonal and orientation-preserving.

The notion of rigid motion is really important in this course because it lets us focus on the important properties of curves and surfaces. For example, the curvature of a curve at a point is invariant under rigid motions. This shows that the curvature is a property that doesn't depend on where the curve is exactly in the space. More generally, the properties we are going to study are these properties that are invariant under rigid motions.

1.2 Topology

In the previous section, we defined the notion of rigid motion, in this section, we define the basic notions of topology required for this course. The most basic notion of topology is the notion of an open set. An open set can be seen as a generalization of an open interval in $\mathbb{R}$. To make this precise, let's first define what disks are in $\mathbb{R}^n$.

Definition. Given $\epsilon > 0$ and $p \in \mathbb{R}^n$, we define the *open disk* (or simply *disk*) as

$$\mathbb{D}_\epsilon(p) = \{q \in \mathbb{R}^n : d_\mathbb{E}(p, q) < \epsilon\}.$$

We also define the closed *closed disk* as

$$\overline{\mathbb{D}_\epsilon(p)} = \{q \in \mathbb{R}^n : d_\mathbb{E}(p, q) \leq \epsilon\}.$$

Even if open and closed disks seem very similar by looking at their definitions, they are in fact very different. To see this difference, suppose that a certain property holds precisely for the points in the open disk $\mathbb{D}_\epsilon(0)$, then we can say that whenever a point p satisfies this property, then there is a small open disk around p such that every point in this smaller disk satisfies this property. However, this is not the case for the closed disk because if we take a point on the boundary, any disk around that point will have elements inside and outside the closed disk. From this, we can now define what it means for a general set to be open.

Definition. A set $U \subset \mathbb{R}^n$ is *open* if for all $p \in U$, there is a $\epsilon > 0$ such that $\mathbb{D}_\epsilon(p) \subset U$. A set $A \subset \mathbb{R}^n$ is *closed* if its complement is open.

The open sets really capture the notion of interior. Every point is at the interior of the set, no point is alone, or on the boundary. The open sets satisfy the following properties.

- Proposition 1.2.1.** (a) Let $\{U_i\}_i$ be an arbitrary collection of open subsets of $\mathbb{R}^n$, then $\bigcup_i U_i$ is an open set.
- (b) Let $\{U_1, \dots, U_n\}$ be a finite collection of open subsets of $\mathbb{R}^n$, then $\bigcap_{i=1}^n U_i$ is an open set.

The notion of disk is also useful to talk about boundedness. In $\mathbb{R}$, a set is bounded if every point satisfies $|x| < M$ for some uniform upper bound M . Equivalently, a set is bounded if it is contained in the interval $(-M, M) = \mathbb{D}_M(0)$. This motivates the following more general definition.

Definition. A set is *bounded* if it is contained in a disk (open or closed).

We see that topology is giving us tools to characterize sets. We defined the notion of open, closed, and bounded sets. There is one last type of sets which will be important for this course that we need to define.

Definition. A set $C \subseteq \mathbb{R}^n$ is *compact* if whenever there is a collection of open sets $\{U_i\}_i$ such that $C \subseteq \bigcup_i U_i$, then there is a subcollection $\{U_1, \dots, U_n\}$ such that $C \subseteq \bigcup_{i=1}^n U_i$.

The notion of compactness may seem very obscure and useless, but it turns out that compact sets are, in a sense, a generalization of finite sets. Moreover, compact sets are very nice to work with for their numerous properties. Thankfully, the following theorem will give us a very useful equivalent definition for compact sets in $\mathbb{R}^n$.

Theorem 1.2.2 (Heine-Borel). A subset $C \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.

As always in mathematics, after adding more structure to a set, we need to define what it means for a function interact well with that new structure. In particular, we need to define what it means for a function from $\mathbb{R}^n$ to $\mathbb{R}^m$ to preserve open sets. This motivates the following definition.

Definition. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *continuous* if $f^{-1}(U) \subseteq \mathbb{R}^n$ is open for all open subsets $U \subseteq \mathbb{R}^m$. If f is continuous, bijective, and its inverse is also continuous, then f is a *homeomorphism*.

Here are important properties of continuous functions:

Proposition 1.2.3. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a continuous function.

- (a) $f^{-1}(A) \subseteq \mathbb{R}^n$ is closed for all closed sets $A \subseteq \mathbb{R}^m$.
- (b) If $C \subseteq \mathbb{R}^n$, then $f(C) \subseteq \mathbb{R}^m$ is compact.

We will not need more topology than what is contained in this section.

1.3 Vector Calculus

In this section, we will state the most important theorem from Vector Calculus: the Inverse Function Theorem. But first, let's define the notion of differentiability for multivariable functions.

Definition. Given a function $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ and a point $a \in U$ where U is open, we say that f is differentiable at p if there is a linear transformation $df_p : \mathbb{R}^n \rightarrow \mathbb{R}^m$, called the *derivative of f at p* , that satisfies

$$\lim_{x \rightarrow p} \frac{\|f(x) - f(p) - df_p(x - p)\|}{\|x - p\|} = 0.$$

The matrix associated to df_p is called the *Jacobian of f at p* and is denoted by $J_f(p)$.

The intuitive way of thinking about the derivative is to view it as the linear approximation of the function at a given point (up to translation). Hence, it should be no surprise that it satisfies most of the properties of the usual derivative:

Proposition 1.3.1. Let $f, g : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ be two functions differentiable at $p \in U$.

- (a) The function f is continuous at p .
- (b) The function λf is differentiable at p , and $d(\lambda f)_p = \lambda df_p$ for all $\lambda \in \mathbb{R}$.
- (c) The function $f + g$ is differentiable at p , and $d(f + g)_p = df_p + dg_p$.
- (d) The function fg is differentiable at p , and $d(fg)_p = g(p)df_p + f(p)dg_p$.
- (e) If $g(p) \neq 0$, then the function f/g is differentiable at p , and

$$d\left(\frac{f}{g}\right)_p = \frac{g(p)df_p - f(p)dg_p}{g(p)^2}.$$

The formulas above should be compared to their one-variable analogue to see that they are really the same. Another important property that really deserves its own theorem is the Chain Rule.

Theorem 1.3.2 (Chain Rule). Let $f : U \subset \mathbb{R}^n \rightarrow \tilde{U} \subset \mathbb{R}^m$ and $g : \tilde{U} \rightarrow \mathbb{R}^k$ be differentiable at $p \in U$ and $f(p) \in \tilde{U}$ respectively, then $g \circ f$ is differentiable at p with

$$d(g \circ f)_p = dg_{f(p)} \circ df_p.$$

In terms of the Jacobian:

$$J_{g \circ f}(p) = J_g(f(p)) \cdot J_f(p).$$

The notion of derivatives lets us classify points in the domain and in the codomain in the following way:

Definition. Let $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a differentiable function. The point $p \in U$ is a *critical point* of f if df_p is not onto $\mathbb{R}^m$ (i.e., $\text{rank}(J_f(p)) < m$). The point $r \in \mathbb{R}^m$ is a *regular value* if it is not the image of a critical point.

In a first multivariable calculus class, we generalize the notion of derivatives by introducing the notion of partial derivatives of a function. The following theorem shows that partial derivatives and Jacobians are related:

Proposition 1.3.3. Let $f = (f_1, \dots, f_m) : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $p \in U$, then all the partial derivatives of f exist and

$$J_f(p) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(p) & \dots & \frac{\partial f_1}{\partial x_n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(p) & \dots & \frac{\partial f_m}{\partial x_n}(p) \end{pmatrix}$$

where the entry in the i th row and j th column is equal to the i th partial derivative of f with respect to x_j . Conversely, if all the partial derivatives of f exist and are continuous at p , then f is differentiable at p .

As it can be seen from the second part of the previous proposition, the fact that all the partial derivatives of a function exist and continuous at a point (or on an open set) is useful, and hence, it deserves its own name.

Definition. A function $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be C^1 if all its partial derivatives exist and are continuous on U . More generally, f is said to be C^2 if all its partial derivatives of the k th order exist and are continuous on U . By convention, we say that f is C^0 if it is continuous. Finally, f is said to be C^∞ , or *smooth*, if it is C^k for all k .

Using this notation, it is clear that any C^{k+1} function is also C^k . This also holds for the case $k = 0$, i.e., every C^1 function is continuous. This follows from the fact that every C^1 function is differentiable, and every differentiable function is continuous. We are now finally able to state the Inverse Function Theorem.

Theorem 1.3.4 (Inverse Function Theorem). Suppose $f : U' \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a C^k function whose derivative at $p \in U'$ is invertible, then there are open sets $U \subseteq U'$ and $V \subseteq \mathbb{R}^n$ such that $p \in U$, $f(U) = V$, and the restriction $f|_U : U \rightarrow V$ is a C^k bijection whose inverse $g : V \rightarrow U$ is C^k with

$$J_g(f(p)) = J_f(p)^{-1}.$$

The one-variable analogue of this theorem is simply the fact that if a function f has a non-zero derivative at a point p in its domain, then the function has an inverse that satisfies $(f^{-1})'(f(p)) = 1/f'(p)$. Writing $y = f(x)$ and using the Leibniz notation gives us the more natural equation

$$\frac{dx}{dy} = \left(\frac{dy}{dx} \right)^{-1}.$$

The Inverse Function Theorem is probably the theorem that we will use the most in what follows.

2 Paths & Curves

2.1 Paths

The goal of this chapter is to define and understand the different properties of curves that are invariant under rigid motions. But first, we need to talk about paths which are simpler objects. In a sense, curves are formed by paths which means that it is very useful to first understand the concept of paths well, and then move on to curves.

Definition. A *path* in $\mathbb{R}^n$ is a continuous function $\gamma : I \rightarrow \mathbb{R}^n$, where $I \subset \mathbb{R}$ is an interval (closed or open, bounded or unbounded). The path is said to be *compact* if I is compact.

Since paths will be used everywhere in this chapter, there are many many new quantities and terminology that need to be defined. First, most paths that we will consider will be differentiable at least once.

Definition. A path $\gamma : I \rightarrow \mathbb{R}^n$ is said to be C^k if the function γ is a C^k function.

Next, it is useful to visualize a path as a function of time. Hence, we will also visualize the first and second derivatives of γ in this way by renaming the derivatives.

Definition. If the path $\gamma : I \rightarrow \mathbb{R}^n$ is C^1 , then we call $\dot{\gamma} : I \rightarrow \mathbb{R}^n$ its *velocity* and $\|\dot{\gamma}\| : I \rightarrow \mathbb{R}$ its *speed*. Moreover, if the path is C^2 , then we call $\ddot{\gamma} : I \rightarrow \mathbb{R}^n$ its *acceleration*.

This lets us define a very important type of path.

Definition. Let $\gamma : I \rightarrow \mathbb{R}^n$ be a C^1 path, then it is said to be *regular* if its velocity is never zero, i.e., if $\dot{\gamma}(t) \neq 0$ for all $t \in I$.

Now that we have defined the main terminology and properties of path, we can start proving that they are all preserved under rigid motions.

Proposition 2.1.1. If $\gamma : I \rightarrow \mathbb{R}^n$ is a (regular) C^k path and $M : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a rigid motion, then $\tilde{\gamma} = M \circ \gamma : I \rightarrow \mathbb{R}^n$ is a (regular) C^k path.

Proof. Write $\gamma = (\gamma_1, \dots, \gamma_n)$ where $\gamma_i : I \rightarrow \mathbb{R}$. Assuming that γ is C^k implies that γ_i is k times differentiable and $\gamma_i^{(k)}$ is continuous. If $M = \vec{a} + T$ with $T = (m_{ij})_{ij}$, then

$$\tilde{\gamma} = \left(a_1 + \sum_{j=1}^n m_{1j} \gamma_j, a_2 + \sum_{j=1}^n m_{2j} \gamma_j, \dots, a_m + \sum_{j=1}^n m_{mj} \gamma_j \right).$$

Since each component of $\tilde{\gamma}$ is a linear combination of k times differentiable functions with continuous k th derivatives, then each component of $\tilde{\gamma}$ is k times differentiable functions with continuous k th derivatives. Thus, $\tilde{\gamma}$ is C^k .

Next, suppose γ is regular and let $t \in I$. By differentiating $\tilde{\gamma}$ component-wise, we get that $\dot{\tilde{\gamma}}(t) = T(\dot{\gamma}(t))$. By our assumption on γ , the vector $\dot{\gamma}(t)$ is non-zero, and hence, $T(\dot{\gamma}(t))$ is also non-zero since T is invertible. Thus, $\tilde{\gamma}$ is regular. ■

By looking at the proof, we see that we didn't need M to be a rigid motion but just a function of the form $T + \vec{a}$ where T is an invertible linear transformation. Hence, we could generalize the previous proposition but we will not since it will not be useful.

Proposition 2.1.2. If $\gamma : I \rightarrow \mathbb{R}^n$ is a C^k path, $M : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a rigid motion and $\tilde{\gamma} = M \circ \gamma : I \rightarrow \mathbb{R}^n$, then $\|\dot{\tilde{\gamma}}\| = \|\dot{\gamma}\|$, i.e., speed is preserved by rigid motions.

Proof. First, write $M = T + \vec{a}$ where T is an orthogonal transformation. As we saw in the proof of the previous proposition, we have $\dot{\tilde{\gamma}}(t) = T(\dot{\gamma}(t))$ for all $t \in I$. Hence, $\|\dot{\tilde{\gamma}}(t)\| = \|T(\dot{\gamma}(t))\|$. But since T is orthogonal, then $\|\dot{\tilde{\gamma}}(t)\| = \|\dot{\gamma}(t)\|$. Therefore, γ and $\tilde{\gamma}$ have the same speed. ■

This time, we had to use the fact that M is a rigid motion. This makes sense geometrically. We will finish this subsection by defining the length of a path.

Definition. Let $\gamma : I \rightarrow \mathbb{R}^n$ be a C^1 path where I is a bounded interval. The *length* of γ is defined as

$$l(\gamma) = \int_I \|\dot{\gamma}(t)\| dt.$$

From what we proved earlier, we can directly prove a useful proposition.

Proposition 2.1.3. The length of a path is preserved under rigid motions.

Proof. Simply notice that the length of a curve only depends on its speed, and that we already proved that speed is preserved under rigid motions. ■

It turns out that as a direct corollary of the fact that length are preserved under rigid motions, we can prove a useful generalization of the triangle inequality in $\mathbb{R}^2$.

Corollary. Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ be a C^1 path with $\gamma(a) = p$ and $\gamma(b) = q$, then

$$\ell(\gamma) \geq d(p, q)$$

with equality holding if and only if γ is a line that never changes direction.

Proof. We can assume that p is at the origin and that q is on the x -axis because the length is unchanged under rigid motions, as well as the fact that γ is C^1 . Once this assumption made, we can write $\gamma = (x, y)$ where x and y are the components of γ . By our assumptions, $x(a) = p$ and $x(b) = q$. Hence:

$$l(\gamma) = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \geq \int_a^b \left|\frac{dx}{dt}\right| dt \geq \left|\int_a^b \frac{dx}{dt} dt\right| = |x(b) - x(a)| = d(p, q).$$

This proves the first part of the theorem. Now, notice that if both quantities were equal, then $(dy/dt)^2 \equiv 0$ and $dx/dt \geq 0$. Equivalently, this means that y is constant and that x is increasing. Since $\gamma(a)$ is on the x -axis and y is constant, then the image of γ is a subset of the x -axis, and hence, a line. Since x is increasing, then this line never changes direction. ■

2.2 Reparametrizations of paths

We usually think of a path as its image. However, many distinct paths have the same image, and hence, many common properties. This is the subject of this section.

Definition. Given two paths $\gamma : I \rightarrow \mathbb{R}^n$ and $\tilde{\gamma} : \tilde{I} \rightarrow \mathbb{R}^n$, then $\tilde{\gamma}$ is a *reparametrization* of γ if there is a homeomorphism $t : \tilde{I} \rightarrow I$ such that $\tilde{\gamma} = \gamma \circ t$.

Intuitively, reparametrizing a path is the same as changing the speed of the path, or even the orientation of the original path. For example, if $\gamma : [a, b] \rightarrow \mathbb{R}^n$ is a path that goes from $\gamma(a)$ to $\gamma(b)$, then $\tilde{\gamma} = \gamma \circ t$ with $t : [a, b] \rightarrow [a, b]$ defined by $t(s) = b + a - s$ is the exact same path as γ , except that it now goes from $\gamma(b)$ to $\gamma(a)$. This motivates the following definition.

Definition. A change of variable t is *orientation-preserving* if it is increasing, and *orientation-reversing* if it is decreasing.

As we did in the previous section, we need to be sure that some essential properties of a path are not lost after a reparametrization.

Proposition 2.2.1. Suppose γ is (regular) C^k path and $t : \tilde{I} \rightarrow I$ is a (regular) C^k bijection, then $\tilde{\gamma} = \gamma \circ t$ is a (regular) C^k reparametrization, i.e., reparametrization preserves the differentiability and regularity.

Proof. By the chain rule, $\tilde{\gamma}$ is a composition of two C^k functions so it is a C^k function. If γ and t are regular, then $\dot{\gamma}(t) \neq 0$ for all $t \in I$ and $t'(s) \neq 0$ for all $s \in \tilde{I}$. By the chain rule, $\dot{\tilde{\gamma}}(s) = t'(s) \cdot \dot{\gamma}(t(s)) \neq 0$ for all $s \in \tilde{I}$ since both factors are non-zero. Thus, $\tilde{\gamma}$ is regular. ■

Since a reparametrization of a path has the same image as the path, it can be guessed that they have the same length. It turns out that we can easily prove this remark under the assumption that the change of variable is C^1 .

Proposition 2.2.2. Suppose $\gamma : I \rightarrow \mathbb{R}^n$ is a C^1 path, $t : \tilde{I} \rightarrow I$ is a C^1 homeomorphism and $\tilde{\gamma} = \gamma \circ t$, then $\ell(\tilde{\gamma}) = \ell(\gamma)$, i.e., reparametrization preserves the length.

Proof. First, notice that t is a homeomorphism between two intervals, hence, t is either increasing or decreasing. Let c be a constant equal to 1 if t is increasing, and -1 if t is decreasing, then $|t'| = c \cdot t'$. Moreover, if t is used as a change of variable in an integral, then multiplying the integral by c will make it stay in the right orientation. It follows that

$$\ell(\tilde{\gamma}) = \int_{\tilde{I}} \|\dot{\tilde{\gamma}}(s)\| ds = c \int_{\tilde{I}} t'(s) \cdot \|\dot{\gamma}(t(s))\| ds = \int_I \|\dot{\gamma}(t)\| dt = \ell(\gamma).$$

■

As we can see from the previous propositions, reparametrizing a path don't change its important properties. There are many situations where reparametrizing a path can be useful because it lets us choose an appropriate parametrization depending on the context. The next proposition motivates this idea.

Proposition 2.2.3. If γ is a regular C^1 surface with constant speed, then its acceleration is perpendicular to its velocity, i.e., $\ddot{\gamma} \perp \dot{\gamma}$.

Proof. Suppose that $\|\dot{\gamma}\| \equiv c$ for some $c > 0$ and notice that

$$\dot{\gamma} \cdot \dot{\gamma} = \|\dot{\gamma}\|^2 = c.$$

Taking the derivative on both sides gives us

$$\frac{d}{dt}(\dot{\gamma} \cdot \dot{\gamma}) = 0.$$

On the other hand, the product rule implies that

$$\frac{d}{dt}(\dot{\gamma} \cdot \dot{\gamma}) = \ddot{\gamma} \cdot \dot{\gamma} + \dot{\gamma} \cdot \ddot{\gamma} = 2(\ddot{\gamma} \cdot \dot{\gamma})$$

Therefore, $\ddot{\gamma} \cdot \dot{\gamma} = 0$ which directly implies that $\ddot{\gamma} \perp \dot{\gamma}$. ■

To motivate this proposition a little bit more, let's first define the tangent and normal vectors at a point to get a more visual intuition of the situation.

Definition (Unit Tangent, Unit Normal). Let $\gamma : I \rightarrow \mathbb{R}^n$ be a regular C^1 path, then the *unit tangent* at $t_0 \in I$ is defined by

$$T(t_0) = \frac{\dot{\gamma}(t_0)}{\|\dot{\gamma}(t_0)\|}.$$

If γ is assumed to be C^2 , the *unit normal* at t_0 is defined by

$$N(t_0) = \frac{\ddot{\gamma}(t_0)^\perp}{\|\ddot{\gamma}(t_0)^\perp\|}.$$

Hence, by the previous proposition, if γ has constant speed, then $\ddot{\gamma}^\perp = \ddot{\gamma}$, and so $N = \ddot{\gamma} / \|\ddot{\gamma}\|$. Thus, constant speed paths are easier to work with and easier to visualize. The question now is if it is always possible to find a reparametrization with constant speed. The next definition and theorem gives us a positive answer.

Definition (Arclength Reparametrization). An *arclength reparametrization* of a regular C^1 path γ is a regular C^1 reparametrization $\tilde{\gamma}$ such that $\|\dot{\tilde{\gamma}}\| \equiv 1$.

Proposition 2.2.4. Let $\gamma : I \rightarrow \mathbb{R}^n$ be a regular C^k path, then there exists a C^k orientation-preserving arclength reparametrization of γ .

Proof. Take an arbitrary $t_0 \in I$ and consider the function $s : I \rightarrow \mathbb{R}$ defined by

$$s(t) = \int_{t_0}^t \|\dot{\gamma}(t)\| dt.$$

Since this function is differentiable and strictly increasing, then its inverse $s^{-1} : s(I) \rightarrow I$ must be differentiable, strictly increasing such that

$$(s^{-1})'(t) = \frac{1}{s'(s^{-1}(t))} = \frac{1}{\|\dot{\gamma}(s^{-1}(t))\|}.$$

Next, define the reparametrization $\tilde{\gamma} = \gamma \circ s^{-1}$, then

$$\|\dot{\tilde{\gamma}}(t)\| = \frac{1}{\|\dot{\gamma}(s^{-1}(t))\|} \|\dot{\gamma}(s^{-1}(t))\| \equiv 1.$$

Therefore, γ has an orientation-preserving arclength reparametrization. ■

2.3 Curves

In this section, we define what a curve is, and look at some basic properties of curves. Following our intuition, we could say that curves and paths are the same, and simply pursue our study of paths. However, it is important to make a clear distinction between paths and curves in the same way that there is a distinction between a function and its graph. This would lead us to define curves as images of paths. Even though this is partly true, we will make the definition of a curve a little bit more general so that the transition to the study of surfaces will be smoother.

Definition (Curve). A connected subset $\mathcal{C} \subset \mathbb{R}^n$ is a (C^k) curve if for all points $p \in \mathcal{C}$, there exists a compact neighborhood N_p of p and a one-to-one compact (regular C^k) parametrized curve $\gamma : I \rightarrow \mathbb{R}^n$ such that $\gamma(I) = \mathcal{C} \cap N_p$.

This definition tells us that a curve is simply a set that locally looks like the image of a compact path, which is not very far from our original intuitive definition.

For example, the unit circle $\mathbb{S}^1 \subset \mathbb{R}^2$ is curve. To prove it rigorously, take a point $p = (x, y) \in \mathbb{S}^1$; if $y \geq 0$, take $N_p = [-1, 1] \times [0, 1]$ and $\gamma : [-1, 1] \rightarrow \mathbb{S}^1 \cap N_p$ defined by $\gamma(t) = (t, \sqrt{1-t^2})$; if $y < 0$, take $N_p = [-1, 1] \times [-1, 0]$ and $\gamma : [-1, 1] \rightarrow \mathbb{S}^1 \cap N_p$ defined by $\gamma(t) = (t, -\sqrt{1-t^2})$. Similarly, the graph of any continuous function $f : I \rightarrow \mathbb{R}$, where I is compact interval, is a curve. This is an interesting example because we have here a curve which is not the image of a single compact path.

A second interesting example is the logarithmic spiral, which is defined as the image of the curve $\gamma : [0, \infty) \rightarrow \mathbb{R}^2$ with $\gamma(t) = e^{-t}(\cos t, \sin t)$. This curve spirals around the origin, which it approaches. It turns out that the curve obtained by adding the origin to the logarithmic spiral is also a curve because it has the following global parametrization: $\gamma(t) : [0, \pi/2] \rightarrow \mathbb{R}^2$ defined by

$$\gamma(t) = \begin{cases} e^{-\tan(t)}(\cos(\tan(t)), \sin(\tan(t))), & t \in [0, \pi/2) \\ (0, 0), & t = \pi/2. \end{cases}$$

Next, as we did for paths, let's define the notion of length of a curve. When $\mathcal{C}$ is a segment with endpoints p and q , we simply define its length as the distance between p and q , and denote it by $\mathcal{L}_1(\mathcal{C})$. In other words:

$$\mathcal{L}_1(\mathcal{C}) = d(p, q).$$

When $\mathcal{C}$ is the image of a C^1 compact path γ , we define the length of $\mathcal{C}$ as the length of γ as it was defined in Section 2.1, and denote it by $\mathcal{L}_2(\mathcal{C})$. In other words:

$$\mathcal{L}_2(\mathcal{C}) = \ell(\gamma).$$

We proved in Section 2.1 that when γ is a parametrization of a segment $\mathcal{C}$, then $\ell(\gamma)$ is the distance between the two endpoints, hence, $\mathcal{L}_1(\mathcal{C}) = \mathcal{L}_2(\mathcal{C})$. This proves that the definitions are compatible. However, we only considered very special cases of curves for the moment. The next definition defines the length of a more class of curves.

Definition (Length of Rectifiable Curves). Let $\mathcal{C}$ be a curve. An *inscribed polygon* in $\mathcal{C}$ is a finite increasing sequence of points $\mathcal{P} = \{p_i\}_{i=0}^N$ of points in $\mathcal{C}$ such that p_0 and

p_N are the endpoints of $\mathcal{C}$. The length of $\mathcal{P}$ is defined as

$$\mathcal{L}(\mathcal{P}) = \sum_{i=0}^N d(p_i, p_{i+1})$$

and its size is defined as

$$|\mathcal{P}| = \max_{0 \leq i \leq N-1} d(p_i, p_{i+1}).$$

A curve is said to be *rectifiable* if there exists a real number L such that every sequence $\{\mathcal{P}_n\}_n$ with $|\mathcal{P}_n| \rightarrow 0$ satisfies

$$\lim_{n \rightarrow \infty} \mathcal{L}(\mathcal{P}_n) = L.$$

We define the length of $\mathcal{C}$ to be this limit L , and denote it by $\mathcal{L}_3(\mathcal{C})$. In other words:

$$\mathcal{L}_3(\mathcal{C}) = \lim_{n \rightarrow \infty} \mathcal{L}(\mathcal{P}_n).$$

We finish this section by stating an important theorem that will make our life easier.

Theorem 2.3.1 (Classification Theorem for Curves). Let $\mathcal{C} \subset \mathbb{R}^n$ be connected, then $\mathcal{C}$ is a (C^k) curve if and only if $\mathcal{C}$ is the image of a (C^k) path $\gamma : I \rightarrow \mathbb{R}^n$ satisfying either of the following:

- (a) γ is one-to-one with a continuous (C^k) inverse;
- (b) $I = \mathbb{R}$, γ is periodic, and $\gamma|_{I'}$ is one-to-one when the length of I' is shorter than the period.

The path γ is said to be a *global parametrization* of $\mathcal{C}$. When the global parametrization is periodic, we call the curve *closed*. We will finish this section by stating two theorems involving closed curves and that are easy to visualize.

Theorem 2.3.2 (Jordan Curve Theorem). Let $\mathcal{C} \subset \mathbb{R}^2$ be a regular closed curve, then $\mathbb{R}^2 \setminus \mathcal{C}$ has exactly two connected components: one bounded (called the inside of the curve), and one unbounded (called the outside of the curve).

Theorem 2.3.3 (Isoperimetric Inequality). Let $\mathcal{C} \subset \mathbb{R}^2$ be a regular closed curve with area A , then

$$4\pi A \leq \ell(\mathcal{C})^2$$

with equality precisely when $\mathcal{C}$ is a circle.

2.4 Curvature

Let's define the notion of curvature. To do this, we will first consider the notion of acceleration of a path. Intuitively, if we think of a path as a trajectory, then the more the path accelerates, the more it *curves*. However, this intuition only works for paths with constant speed because our intuition breaks when the path is a straightline accelerating in the same direction. In this example, the acceleration is non-zero even though the path is never *curved*. This motivates our first definition of curvature for paths.

Definition (Curvature (Paths)). Let $\gamma : I \rightarrow \mathbb{R}^n$ be a regular C^2 path. The curvature of γ at $t_0 \in I$ is defined as

$$\kappa_\gamma(t_0) = \|\ddot{\tilde{\gamma}}(s_0)\|$$

where $\tilde{\gamma} = \gamma \circ t : \tilde{I} \rightarrow \mathbb{R}^n$ is an orientation preserving arclength reparametrization of γ , and $t(s_0) = t_0$.

This is well-defined and independent of the choice of reparametrization by what was proved in Assignment 1. Now that we have a well-defined notion of curvature for paths, we get the following definition of curvature for curves.

Definition (Curvature (Curves)). Given a regular C^2 curve $\mathcal{C} \subset \mathbb{R}^n$, there exists a global regular C^2 para-metrization of $\mathcal{C}$, $\gamma : I \rightarrow \mathbb{R}^n$. For a point $p \in \mathcal{C}$, define the curvature of $\mathcal{C}$ at p as $\kappa_\gamma(t_0)$ where $\gamma(t_0) = p$. If we denote the curvature of $\mathcal{C}$ at p by $\kappa_{\mathcal{C}}(p)$, we get that $\kappa_{\mathcal{C}}(p) = \kappa_\gamma(\gamma^{-1}(p))$.

Again, the same problem arises: is it well-defined? How can we be sure that this doesn't depend on the choice of global parametrization of $\mathcal{C}$? This is in fact easy to show that this is well-defined because any two global parametrizations of $\mathcal{C}$ are reparametrizations of each other. Hence, since the curvature is well-defined for paths, then the curvature is the same for both global parametrizations. Thus, the curvature is well-defined for curves.

Next, to motivate the importance of the curvature, let's now show that the curvature is invariant under rigid motions.

Proposition 2.4.1. Let $\mathcal{C} \subset \mathbb{R}^n$ be a regular C^2 curve and M a rigid motion of $\mathbb{R}^n$, then $\kappa_{M(\mathcal{C})} = \kappa_{\mathcal{C}}$.

Proof. Since everything is well-defined, it suffices to prove that $\kappa_{M \circ \gamma} = \kappa_\gamma$ for some regular C^2 unit-speed path γ . If we write $M = T + \vec{a}$ where T is orthogonal, and $\tilde{\gamma} = M \circ \gamma$, then we already showed that $\tilde{\gamma}$ is a regular C^2 unit-speed path and $T \circ \ddot{\tilde{\gamma}} = \ddot{\gamma}$. Therefore,

$$\kappa_{M \circ \gamma} = \|T \circ \ddot{\tilde{\gamma}}\| = \|\ddot{\gamma}\| = \kappa_\gamma.$$

■

No, the only problem with the curvature is that its definition makes it hard to compute. Fortunately, there is an alternative way of computing the curvature of a path at a point with having to find an orientation preserving arclength reparametrization (which might be hard or time consuming).

Proposition 2.4.2. Let $\gamma : I \rightarrow \mathbb{R}^n$ be a regular C^2 path, then

$$\kappa_\gamma = \frac{1}{\|\dot{\gamma}\|^2} \left\| \ddot{\gamma} - \frac{\ddot{\gamma} \cdot \dot{\gamma}}{\dot{\gamma} \cdot \dot{\gamma}} \dot{\gamma} \right\| = \frac{\|\ddot{\gamma}^\perp\|}{\|\dot{\gamma}\|^2}$$

where $\ddot{\gamma}^\perp$ is the component of $\ddot{\gamma}$ which is perpendicular to $\dot{\gamma}$.

Proof. Let $\tilde{\gamma} = \gamma \circ t$ be an orientation preserving arclength reparametrization of γ and denote t^{-1} by s . By the rules of differentiation:

$$\begin{aligned}\tilde{\gamma} = \gamma \circ t &\implies \dot{\tilde{\gamma}} = (\dot{\gamma} \circ t) \cdot t' \\ &\implies \ddot{\tilde{\gamma}} = (\ddot{\gamma} \circ t) \cdot (t')^2 + (\dot{\gamma} \circ t) \cdot t''.\end{aligned}$$

The full proof is just by direct calculation of $\|\ddot{\tilde{\gamma}} \circ s\|$ [should be completed]. $\blacksquare$

For example, if we take the curve $\gamma(t) = (t, t^2)$, then $\dot{\gamma}(t) = (1, 2t)$ and $\ddot{\gamma}(t) = (0, 2)$. Hence,

$$\|\ddot{\gamma}^\perp(t)\| = \left\| \begin{pmatrix} 0 \\ 2 \end{pmatrix} - \frac{\begin{pmatrix} 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2t \end{pmatrix}}{\begin{pmatrix} 1 \\ 2t \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2t \end{pmatrix}} \begin{pmatrix} 1 \\ 2t \end{pmatrix} \right\| = \left\| \begin{pmatrix} -4t/(1+4t^2) & 2/(1+4t^2) \end{pmatrix} \right\| = \frac{2}{\sqrt{1+4t^2}}$$

which gives us

$$\kappa_\gamma(t) = \frac{2}{(1+4t^2)^{3/2}}.$$

It turns out that there is another way of thinking about the curvature of a curve at a point. Intuitively, the curvature of a circle is constant along the circle and so it only depends on its radius. Moreover, the curvature of a circle is (intuitively) smaller as the radius gets bigger because if the radius is very large, the circle becomes nearly indistinguishable from a straightline which has curvature zero (following our intuition). Thus, we can define the radius of a circle of radius r to be $1/r$. Finally, the curvature of a curve at a point p is simply the the inverse of the radius of the circle that best approximates the curve at p .

This new way of thinking about the curvature is more visual and is consistent with the first definition we gave. To see this, let C be a circle of radius r and suppose that it is centered at the origin, then we can pick the parametrization $\gamma(\theta) = r(\cos \theta, \sin \theta)$ which has constant speed. Thus, the curvature is given by

$$\kappa_\gamma(\theta) = \frac{\|\ddot{\gamma}(\theta)\|}{\|\dot{\gamma}(\theta)\|^2} = \frac{r \|(-\cos \theta, -\sin \theta)\|}{r^2 \|(-\sin \theta, \cos \theta)\|} = \frac{1}{r}.$$

The previous discussion of the alternative purely visual way of defining the curvature motivates the following definition.

Definition (Osculating Plane, Osculating Circle, Evolute). Let $\gamma : I \rightarrow \mathbb{R}^n$ be a regular C^2 path with non-zero curvature at $t \in I$, then the unit tangent and unit normal are linearly independent. Define the *osculating place* of γ at t to be the plane containing $\gamma(t)$ and spanned by the vectors $T(t)$ and $N(t)$.

Define the *osculating circle* to be the circle in the osculating plane of radius $1/\kappa_\gamma(t)$ and center $\lambda(t) = \gamma(t) + N(t)/\kappa_\gamma(t)$. The path $\lambda(t)$ is called the *evolute* of γ .

In the same way that we can think of the tangent line of a curve at a point as the straightline approximating the curve at that point, the osculating circle really is the "circle approximation" of the curve at that same point. Hence, the curvature can be seen as an analogue of the derivative in terms of the local information it gives about a curve at a point.

2.5 Plane Paths

In this chapter, we will consider plane paths only because it will be hard to generalize some of the tools we are going to introduce.

The curvature is independent of the orientation of a curve. More precisely, if $\tilde{\gamma} = \gamma \circ t$ is a reparametrization of γ such that t is orientation-reversing, we will still have $\kappa_{\tilde{\gamma}} = \kappa_{\gamma}$; the curvature doesn't depend on the nature of the change of variable. It turns out that the definition of the curvature can be slightly modified to encapsulate information about the orientation as well.

To do this, consider a regular C^2 path γ and a point t_0 in its domain. Since the curvature is invariant under rigid motions, we can assume that $\gamma(t_0)$ is the origin of the plane and that $T(t_0)$ is the standard vector $\vec{e}_1$. In this setup, we get that the vector $\ddot{\gamma}(t_0)^\perp / \|\dot{\gamma}(t_0)\|^2$ (whose norm is $\kappa_{\gamma}(t_0)$) is perpendicular to $\vec{e}_2$. Hence, we get that

$$\frac{\ddot{\gamma}(t_0)^\perp}{\|\dot{\gamma}(t_0)\|^2} = \alpha \cdot \vec{e}_2$$

for some $\alpha \in \mathbb{R}$. We define the "signed curvature" of γ as this constant α . To get a more precise definition, we need to solve for α . This gives us the following definition.

Definition (Signed Curvature). Let $\gamma : I \rightarrow \mathbb{R}^2$ be a regular C^2 path, and fix $t \in I$. The *signed curvature* of γ at t is defined as

$$\kappa_{\gamma}^{\pm}(t) = \frac{\ddot{\gamma}(t) \cdot \dot{\gamma}(t)^*}{\|\dot{\gamma}(t)\|^3}$$

where $(x, y)^* = (-y, x)$.

The next proposition follows directly from the definition.

Proposition 2.5.1. Let $\gamma : I \rightarrow \mathbb{R}^2$ be a regular C^2 path of the form $\gamma = (x, y)$, then

$$\kappa_{\gamma}^{\pm} = \frac{\dot{x}\ddot{y} - \ddot{x}\dot{y}}{(x^2 + y^2)^{3/2}}.$$

From this definition, we see that the signed curvature is positive when the path is turning counterclockwise, and negative when it is turning clockwise. Similarly, we can already prove two properties of the signed curvature which confirm our visual intuition of this new object.

Proposition 2.5.2. The signed curvature satisfies the following properties:

- (a) $|\kappa^{\pm}| = \kappa$;
- (b) $\kappa_{\gamma_-}^{\pm} = -\kappa_{\gamma}^{\pm}$.

The proofs of both properties are easy. The second property is simple and easy to prove but it is of great important because it proves that the signed curvature is not well-defined for curves. This comes from the fact that a curve don't have an orientation until we chose one. For example, the unit circle can be parametrized by a clockwise rotation, as well as a counterclockwise rotation. In the first case, the signed curvature would be negative, and in the second, positive. However, beside this ambiguity in the sign, the absolute value of the signed curvature is well-defined since it is simply the curvature. Thus, we can say that the signed curvature is well-defined for oriented curves.

Notice that using the signed curvature, we can rewrite the evolute in the following simpler way:

$$\lambda(t) = \gamma(t) + \frac{1}{\kappa_{\gamma}^{\pm}(t)} \frac{\dot{\gamma}(t)^*}{\|\dot{\gamma}(t)\|}.$$

Finally, there is one last function that needs to be defined in this section to complete our tools for the study of the curvature: the *angle function*. To motivate this idea, let γ be a path and consider the unit tangent $T(t)$ as a function of t . As time varies, the vector $T(t)$ varies on the unit circle. This motivates the existence of a function $\theta(t)$ such that $T(t) = (\cos \theta(t), \sin \theta(t))$. However, the construction of this function might not be clear. The key is to notice the following link between the function $\theta(t)$ and the signed curvature: if $\theta(t)$ has slow variations, then the curvature is small; if $\theta(t)$ varies quickly, then the magnitude of the curvature is high. This hints towards the following relation: $\theta'(t) = \kappa_{\gamma}^{\pm}(t)$. This is confirmed by the following derivation: if $T(t) = (\cos \theta(t), \sin \theta(t))$ and γ is assumed to have unit-speed, then

$$\dot{\gamma}(t) = (\cos \theta(t), \sin \theta(t)).$$

Taking the derivative on both sides:

$$\ddot{\gamma}(t) = \theta'(t)(-\sin \theta(t), \cos \theta(t)).$$

Taking the dot product with $\dot{\gamma}(t)^* = (-\sin \theta(t), \cos \theta(t))$ gives us

$$\kappa_{\gamma}^{\pm}(t) = \theta'(t).$$

Thus, we get

$$\theta(t) = \int_{t_0}^t \kappa_{\gamma}^{\pm}(u) du + \theta(t_0)$$

for some $t_0 \in I$. Finally, for a general path γ which is not necessarily unit-speed, it suffices to make a change of variable. Therefore, we can now rigorously define the angle function.

Definition (Angle Function). Let $\gamma : I \rightarrow \mathbb{R}^2$ be a regular C^2 path, $t_0 \in I$ and $\theta_0 \in \mathbb{R}$ such that $T(t_0) = (\cos \theta_0, \sin \theta_0)$, then the *angle function* $\theta : I \rightarrow \mathbb{R}$ is defined by

$$\theta(t) = \int_{t_0}^t \kappa_{\gamma}^{\pm}(u) \|\dot{\gamma}(u)\| du + \theta_0.$$

Proposition 2.5.3. Let $\gamma : I \rightarrow \mathbb{R}^2$ be a regular C^2 path, then

$$T(t) = (\cos \theta(t), \sin \theta(t))$$

for all $t \in I$.

Proof. Since the unit tangent function is invariant under orientation-preserving reparametrization, we can assume that γ is parametrized by arclength. In that case, $T(t) = \dot{\gamma}(t)$ and $\theta'(t) = \kappa_{\gamma}^{\pm}(t)$. If we define the function $v(t) = (\cos \theta(t), \sin \theta(t))$, then it satisfies the following relations:

$$v'(t) = \kappa_{\gamma}^{\pm}(t)v(t)^*, \quad v(t_0) = T(t_0).$$

Next, notice that $T(t)$ satisfies the same two relations. Thus, since the relations define an IVP, the Picard-Lindelöf Theorem lets us conclude that $v(t) = T(t)$ for all $t \in I$. ■

The following theorem is easy to state, useful, and it shows that the angle function behaves according to our intuition. In other words, this theorem makes sense when we think about it.

Theorem 2.5.4 (Hopf's Umlaufsatz). Let $\mathcal{C} \subset \mathbb{R}^2$ be a regular C^2 closed curve with global parametrization $\gamma : \mathbb{R} \rightarrow \mathbb{R}^2$ with period $[a, b]$, then $|\theta(b) - \theta(a)| = 2\pi$.

Unfortunately, the proof uses homotopy theory and so I decided to not include it in these notes. Basically, this theorem says that a closed curve forms a loop. Notice that we can define the clockwise and counter-clockwise orientations as a consequence of this theorem: If $\theta(b) - \theta(a) = 2\pi$, we say that γ has the counter-clockwise orientation; if $\theta(b) - \theta(a) = -2\pi$, we say that γ has the clockwise orientation.

We can now finish this section with the theorem that will prove the importance of the notion of curvature: the Fundamental Theorem of Plane Paths.

Theorem 2.5.5 (Fundamental Theorem of Plane Paths). Let $I \subset \mathbb{R}$ be an interval, $s_0 \in I$, $2 \leq k \leq \infty$, and $K : I \rightarrow \mathbb{R}$ be a C^{k-2} function, then for all $\vec{p} \in \mathbb{R}^2$ and $\theta_0 \in \mathbb{R}$, there exists a unique regular C^k path $\gamma : I \rightarrow \mathbb{R}^2$ parametrized by arclength such that

- (a) $\kappa_{\gamma}^{\pm} = K$;
- (b) $\dot{\gamma}(s_0) = (\cos \theta_0, \sin \theta_0)$;
- (c) $\gamma(s_0) = \vec{p}$.

Proof. First, define the function θ as follows:

$$\theta(t) = \int_{s_0}^t K(u)du + \theta_0.$$

Next, define the path γ by

$$\gamma(t) = \left(\int_{s_0}^t \cos \theta(u)du, \int_{s_0}^t \sin \theta(u)du \right) + \vec{p}.$$

Finally, notice that the function γ is a regular C^k path parametrized by arclength that satisfies the three conditions above.

To prove the uniqueness, suppose that $\gamma_1 : I \rightarrow \mathbb{R}^2$ is another path satisfying the three conditions. Since γ and γ_1 have the same signed curvature, then they have the same angle function. It follows that $\dot{\gamma}$ and $\dot{\gamma}_1$ are equal. Finally, by the fact that $\gamma(s_0) = \gamma_1(s_0) = \vec{p}$, we get that $\gamma = \gamma_1$. Therefore, γ exists and it is unique. ■

This theorem states that unit-speed paths are entirely determined by their signed curvature. For example, this theorem lets us prove easily that the paths with constant non-zero signed curvature are precisely the circles.

2.6 Space paths

Let's move to dimension 3. If γ is a regular C^2 path with positive curvature, then the tangent and normal vectors are always defined. Since we are working in $\mathbb{R}^3$, we can define a third vector to form a basis.

Definition (Binormal Vector). Let $\gamma : I \rightarrow \mathbb{R}^3$ be a regular C^2 path with $\kappa > 0$. We define the *binormal* vector $B(t) = T(t) \times N(t) \neq \vec{0}$. These three vectors form a basis which we call the *Frenet Frame*.

We are going to be interested in the dynamics of this frame. Suppose that γ is now C^3 and parametrized by arclength, then

$$\frac{dT}{ds} = \dot{\gamma} = \kappa N.$$

Next, by the product rule and the previous equation, we have that $\frac{dB}{ds} = T \times \frac{dN}{ds}$. Since the cross product of two vectors is orthogonal to both vectors, we get that $\frac{dB}{ds} \perp T$. Similarly, since $\|B\| \equiv 1$, we get that $\frac{dB}{ds} \perp B$. Therefore, we must have that $\frac{dB}{ds} = \alpha N$. If we define the torsion τ as the negative of this constant, we get the equation

$$\frac{dB}{ds} = -\tau N.$$

The function τ is called the *torsion*. Next, we already know $\frac{dN}{ds} \perp N = 0$ so $\frac{dN}{ds}$ is only composed of T and B . To find the coefficients, start from the equation $T \cdot N = 0$ and differentiate both sides to get $T \cdot \frac{dN}{ds} = -\kappa$. Similarly, start from the equation $B \cdot N = 0$ and differentiate both sides to get $B \cdot \frac{dN}{ds} = \tau$. Thus,

$$\frac{dN}{ds} = -\kappa T + \tau N.$$

These three equations are called the *Frenet Equations*. They can be summarized as follows.

Theorem 2.6.1 (Frenet Equations). Let $\gamma : I \rightarrow \mathbb{R}^3$ be a regular C^3 path parametrized by arclength, then

$$\begin{pmatrix} \dot{T} \\ \dot{N} \\ \dot{B} \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

which is a 1st order linear ODE.

It turns out that now have all we need to state and prove an analogue of the Fundamental Theorem of Plane Paths, but for space paths.

Theorem 2.6.2 (Fundamental Theorem of Space Paths). Let $I \subseteq \mathbb{R}$ be an interval with basepoint $s_0 \in I$. Suppose $\tau : I \rightarrow \mathbb{R}$ is a C^{k-3} function, and $\kappa : I \rightarrow \mathbb{R}_{>0}$ is a C^{k-2} function. Then for any initial position p_0 , initial velocity v_0 , and initial normal n_0 in $\mathbb{R}^3$ such that $\|v_0\| = \|n_0\| = 1$ and $v_0 \cdot n_0 = 1$, there is a unique regular C^k path $\gamma : I \rightarrow \mathbb{R}^3$ parametrized by arclength and satisfying the following conditions:

$$\kappa_\gamma = \kappa, \quad \tau_\gamma = \tau, \quad \gamma(s_0) = p_0, \quad \dot{\gamma}(s_0) = v_0, \quad \ddot{\gamma}(s_0)/\|\ddot{\gamma}(s_0)\| = n_0.$$

Proof. If we look at the Frenet Equations and the given values in the statement of theorem, we can recognize that we have a 1st order linear IVP. By the Picard-Lindelöf Theorem, there is a unique solution $T, N, B : I \rightarrow \mathbb{R}^3$. Let's find γ such that T, N, B are the corresponding Frenet frame of γ . To do this, let's show that they form an orthonormal basis. By the Frenet Equations,

$$\begin{aligned} \frac{d}{ds}(T \cdot N) &= \kappa(N \cdot N) - \kappa(T \cdot T) + \tau(T \cdot B), \\ \frac{d}{ds}(T \cdot N) &= \kappa(N \cdot B) - \tau(T \cdot N), \\ \frac{d}{ds}(N \cdot B) &= -\kappa(T \cdot B) + \kappa(B \cdot B) - \tau(N \cdot N), \\ \frac{d}{ds}(T \cdot T) &= 2\kappa(T \cdot T), \\ \frac{d}{ds}(N \cdot N) &= -2\kappa(T \cdot N) + 2\tau(N \cdot B), \\ \frac{d}{ds}(B \cdot B) &= -2\tau(N \cdot B). \end{aligned}$$

This is a system of six first order ODEs with initial values $0, 0, 0, 1, 1, 1$ which has a unique solution again. Since the constant functions $0, 0, 0, 1, 1, 1$ also solve this IVP, then by uniqueness,

$$T \cdot N = T \cdot B = N \cdot B \equiv 0, \quad T \cdot T = N \cdot N = B \cdot B \equiv 1.$$

Therefore, T, N, B form an orthogonal basis. Finally, simply define $\gamma : I \rightarrow \mathbb{R}^3$ by

$$\gamma(s) = \int_{s_0}^s T(u) du + \vec{p}.$$

■

Before ending this section, let's talk a little bit more about the torsion. As we can see from the Fundamental Theorem of Space Paths, it is clear that the torsion is as important as the curvature for characterizing a path in $\mathbb{R}^3$. For a general path (not necessarily parametrized by arclength), we can give the following definition of the torsion.

Definition (Torsion). Let $\gamma : I \rightarrow \mathbb{R}^3$ be a regular C^3 with $\kappa > 0$, then the *torsion* $\tau : I \rightarrow \mathbb{R}$ is defined by

$$\tau = \tilde{\tau} \circ s$$

where $s : I \rightarrow \tilde{I}$ is an orientation-preserving change of parameter from γ to its arclength reparametrization $\tilde{\gamma}$, and $\tilde{\tau}$ is the torsion of $\tilde{\gamma}$ as defined above.

As always, this might seem complicated to compute but it turns out that there are simpler formulas, as the one given by the following proposition.

Proposition 2.6.3. Let $\gamma : I \rightarrow \mathbb{R}^3$ be a regular C^3 path with non-zero curvature, then

$$\kappa = \frac{\|\dot{\gamma} \times \ddot{\gamma}\|}{\|\dot{\gamma}\|^3} \quad \text{and} \quad \tau = \frac{\dot{\gamma} \times \ddot{\gamma}}{\|\dot{\gamma} \times \ddot{\gamma}\|^2} \cdot \ddot{\gamma}.$$

The proof is not very enlightening so I will skip it. It is on this note that this chapter ends. The goal was to develop all the tools which are useful for the analysis of a path or a curve. As it can be noticed, the notion of a curve was used a few times but we mostly expressed our theorems in terms of paths. This is because paths are easier to work with and thanks to the Classification Theorems for Curves. Unfortunately, no such theorem exists for surfaces.

3 Surfaces

3.1 Definitions and Basic Properties

In this chapter, we will introduce the notion of a surface and develop all the tools that are useful for describing and characterizing a surface. Unfortunately, working with surfaces is harder than working with curves because surfaces don't necessarily have global parametrizations.

First, we define a surface in the same way we defined a curve. Intuitively, we would want a surface to be a subset $\mathcal{S}$ of $\mathbb{R}^n$ such that at each point $p \in \mathcal{S}$, the surface looks like the image of a local parametrization $\varphi : U \subset \mathbb{R}^2 \rightarrow \mathcal{S}$. Now, what would it mean for a surface to be regular? The local parametrization is now a multivariable function so we can't really write $\varphi' \neq 0$. Instead, we now want $d\varphi_q$ to be one-to-one for all q in the domain of φ . Hence, we can now give the rigorous definition.

Definition (Surface). A subset $\mathcal{S}$ of $\mathbb{R}^n$ is a regular C^k surface ($2 \leq n, 1 \leq k \leq \infty$) if for all points $p \in \mathcal{S}$, there are open subsets $U \subseteq \mathbb{R}^2$, $p \in V \subseteq \mathbb{R}^n$, and a C^k homeomorphism $\varphi : U \rightarrow V \cap \mathcal{S} \subseteq \mathbb{R}^n$ such that $d\varphi_q : \mathbb{R}^2 \rightarrow \mathbb{R}^n$ is one-to-one (has rank 2) for all $q \in U$.

We defined a surface by adapting the definition of a curve. It should be clear that both definitions are nearly the same. It turns out that we can define a general mathematical objects such that curves and surfaces are just special cases of this new object. This object, called a *manifold*, is obtained by generalizing the definition of a curve and a surface.

Definition (Manifold). A subset M of $\mathbb{R}^n$ is a regular C^k m -manifold if for all points $p \in M$, there are open subsets $U \subseteq \mathbb{R}^m$, $p \in V \subseteq \mathbb{R}^n$, and a C^k homeomorphism $\varphi : U \rightarrow V \cap M \subseteq \mathbb{R}^n$ such that $d\varphi_q : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is has rank m for all $q \in U$.

Hence, a curve is a 1-manifold, a surface is a 2-manifold. It should be noticed that in the previous chapter, the definition of a curve used compact neighborhoods instead of open sets, and needed to be connected. This is not the case anymore. From now on, a curve is 1-manifold.

Before proving some basic properties of surfaces, let's just mention that there exists a mathematical object analog of paths but for surfaces. It is useful to know and it is the right place to define it.

Definition (Surface Patch / Parametrized Surface). A regular C^k function $\varphi : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$ where U is open is to be a *surface patch*, or sometimes a *parametrized surface*, if $d\varphi_q$ is one-to-one for all $q \in U$.

Notice that a surface patch is not required to be a homeomorphism.

For now, we don't have example of surfaces because it would take too long to prove that a particular set is a surface using the definition, and do it one by one for each example we come up with. The next theorems will give us short cuts to generate surfaces.

Proposition 3.1.1 (Graphs are surfaces). If $U \subseteq \mathbb{R}^2$ is open and $f : U \rightarrow \mathbb{R}$ is a C^k function, then the graph of f in $\mathbb{R}^3$ is a regular C^k surface.

Proof. Use $V = \mathbb{R}^3$, $S = \text{graph}(f) \subseteq \mathbb{R}^3$ and $\varphi : U \rightarrow \mathbb{R}^3 : (x, y) \mapsto (x, y, f(x, y))$. Note that φ is a C^k bijection from U to S with a continuous inverse $\pi : (x, y, z) \mapsto (x, y)$. The jacobian of φ is

$$J_\varphi = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ f_x & f_y \end{pmatrix}$$

which has rank 2 for any input it is given (the two columns are always linearly independant, for any the value for the partial derivatives). Therefore, $\text{graph}(f)$ is a regular C^k surface. ■

It directly follows that the set defined by $z = x^2 + y^2$ is a regular C^∞ surface. The next theorem generalizes this result even more.

Proposition 3.1.2 (Level sets are surfaces). Suppose $\tilde{V} \subseteq \mathbb{R}^3$ is open, $F : \tilde{V} \rightarrow \mathbb{R}$ is a C^k function, and $a \in \mathbb{R}$ is a regular value of f , then $F^{-1}(a)$ is a regular C^k surface.

Proof. Let $p \in F^{-1}(a)$, then $\nabla F(p) = (F_x(p), F_y(p), F_z(p)) \neq 0$ since a is a regular value. Thus, one of $F_x(p), F_y(p), F_z(p)$ must be non-zero. Without loss of generality, we can assume that $F_z(p) \neq 0$. Define the function $G : \tilde{V} \rightarrow \mathbb{R}^3$ by $G(x, y, z) = (x, y, F(x, y, z))$ and compute its Jacobian at p :

$$J_G(p) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ F_x(p) & F_y(p) & F_z(p) \end{pmatrix}$$

Since $\det J_g(p) = F_z(p) \neq 0$, the Inverse Function Theorem states that there is a C^k homeomorphism $H : \hat{U} \rightarrow V \subseteq \tilde{V}$ which is inverse to G . Let U be the projection on $\mathbb{R}^2$ of the set $\hat{S} = \{z = a\}$ and define the homomorphism $\psi : U \rightarrow \hat{S} \cap \hat{U} : (x, y) \mapsto (x, y, a)$. Finally, define the C^k homeomorphism $\varphi = H \circ \psi : U \rightarrow F^{-1}(a) \cap V$. Its Jacobian is

$$J_\varphi(q) = J_H(\psi(q))J_\psi(q) = J_G(H(\psi(q)))^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Since the first matrix is invertible and the second matrix as rank 2, then their composition has rank 2. Therefore, $F^{-1}(a)$ is a regular C^k surface. ■

It follows that the set $x^2 + y^2 + z^2 = 1$ is a regular smooth surface. Some other examples are the hyperboloid with 2-sheets given by the equation $x^2 + y^2 - z^2 = -1$, and the hyperboloid 1-sheet given by the equation $x^2 + y^2 - z^2 = 1$. These examples show that surfaces can be disconnected.

The surfaces generated by the previous theorems might be very special cases in the large class of regular surfaces. It turns out that this is not entirely true in the sense that every surface is locally the graph of a function. This is the content of the next theorem.

Theorem 3.1.3 (Surfaces are local graphs). A subset $S \subseteq \mathbb{R}^3$ is a regular C^k surface if and only if for all points $p \in S$, there are open subsets $U \subseteq \mathbb{R}^2$, $V \subseteq \mathbb{R}^3$ with $p \in V$ such that $V \cap S$ is the graph of a C^k function $U \rightarrow \mathbb{R}$, i.e., it can be described by an equation of the form $z = f(x, y)$, $y = g(x, z)$, or $x = h(y, z)$.

Proof. Being a surface is a local property so if the subset is locally a graph, then it must be a surface since graphs are surfaces. This proves the reverse implication.

For the direct implication, let $p \in S$, then there are open sets $\tilde{U} \subset \mathbb{R}^2$, $p \in \tilde{V} \subset \mathbb{R}^3$ and a C^k homeomorphism $\varphi : \tilde{U} \rightarrow \mathbb{R}^3$ such that $\varphi(\tilde{U}) = \tilde{V} \cap S$ and $J_\varphi(q)$ has rank 2 for all $q \in \tilde{U}$. Define q such that $\varphi(q) = p$ and assume without loss of generality that the first two rows of $J_\varphi(q)$ are linearly independent. Write $\varphi(s, t) = (x(s, t), y(s, t), z(s, t))$ and define the C^k function $\psi : \tilde{U} \rightarrow \mathbb{R}^2$ by $\psi(s, t) = (x(s, t), y(s, t))$. Since $J_\psi(q)$ is precisely the matrix composed of the first two rows of $J_\varphi(q)$, then it is invertible by our assumption. Thus, by the Inverse Function Theorem, there is an open neighborhood $U \subset \tilde{U}$ of q and an open set $W \subset \mathbb{R}^2$ such that $\psi(U) = W$ and $\psi|_U$ is a C^k homeomorphism with inverse $\Psi : W \rightarrow U$. Consider

$$(\varphi \circ \Psi)(x, y) = (x, y, (z \circ \Psi)(x, y)).$$

Finally, since φ is a homeomorphism, then there exists an open set $V \subset \tilde{V}$ such that $V \cap S = (\varphi \circ \Psi)(W)$. Thus, $V \cap S$ is the graph of $z \circ \Psi : W \rightarrow \mathbb{R}$. ■

This theorem gives us a useful characterization of surfaces that will let us prove that some subsets of $\mathbb{R}^3$ are not surfaces. Suppose that $z^2 = x^2 + y^2$ is a surface, then by the previous theorem, this implies that it is the graph of a function around $(0, 0, 0)$. In other words, there are open subsets $U \subset \mathbb{R}^2$ and $V \subset \mathbb{R}^3$ with $(0, 0, 0) \in V$ such that $V \cap \{z^2 = x^2 + y^2\}$ can be described as $z = f(x, y)$, $y = f(x, z)$ or $x = f(y, z)$. However, in all possible cases, we get that the function must be multivalued. Therefore, by contradiction, it is not a surface. similarly, the set $\{z = \sqrt{x^2 + y^2}\}$ because it is not the graph of a differentiable function around the origin.

3.2 Orientation

Now that we are done defining and stating the basic results on surfaces, we will focus in this section on the orientation of a surface (which will be different from the orientation of a path). We start by introducing the tangent space.

Definition (Tangent Space (Surface Patch)). Let $\varphi : U \rightarrow \mathbb{R}^3$ be a C^1 surface patch, and fix $q \in U$. The *tangent space* $T_q\varphi$ to φ at q is the image of the derivative of φ at q .

We would like to define the tangent space of a surface at a point. The most intuitive way to do it would be to define it as the tangent space of a surface patch around that point. The problem is that surface patches at a point are not unique and so it would be undefined. Fortunately, it turns out that the tangent space at a point is independent of the choice of surface patch.

Proposition 3.2.1. If $\tilde{\varphi} : \tilde{U} \rightarrow \mathbb{R}^3$ is a regular C^k reparametrization of $\varphi : U \rightarrow \mathbb{R}^3$ using the change of parameters $h : \tilde{U} \rightarrow U$, then $T_{\tilde{q}}\tilde{\varphi} = T_q\varphi$ where $q = h(\tilde{q})$.

Proof. By definition, $T_{\tilde{q}}\tilde{\varphi} = d\tilde{\varphi}_{\tilde{q}}(\mathbb{R}^2)$. By the chain rule:

$$d\tilde{\varphi}_{\tilde{q}}(\mathbb{R}^2) = (d\varphi_{h(\tilde{q})} \circ dh_{\tilde{q}})(\mathbb{R}^2).$$

But since $h : \tilde{U} \subseteq \mathbb{R}^2 \rightarrow U \subseteq \mathbb{R}^2$ is invertible, then $dh_{\tilde{q}}(\mathbb{R}^2) = \mathbb{R}^2$. Therefore,

$$T_{\tilde{q}}\tilde{\varphi} = d\varphi_{h(\tilde{q})}(dh_{\tilde{q}}(\mathbb{R}^2)) = T_q\varphi.$$

■

We can now define rigorously the tangent space of a general surface at a point.

Definition (Tangent Space (Surface)). Let $\mathcal{S}$ be a surface and $p \in \mathcal{S}$. The *tangent space* of $\mathcal{S}$ at p is defined as the image of any surface patch of $\mathcal{S}$ around p . It is denoted by $T_p\mathcal{S}$.

It can be showed that the tangent space at a point p in a curve in $\mathcal{C} \subseteq S$ is a one dimensional subspace of the tangent space at p of the surface S .

To illustrate the tangent space with a concrete example, let $\mathbb{S}^2$ be the unit sphere in $\mathbb{R}^3$ centered at 0. Let $p = (1, 1, 1)/\sqrt{3} \in \mathbb{S}^2$. We can parametrize the top hemisphere of this surface by the graph of the function $z = \sqrt{1 - x^2 - y^2}$, hence, $\varphi_N : U \rightarrow \mathbb{S}^2$ is given by $\varphi(x, y) = (x, y, \sqrt{1 - x^2 - y^2})$ where U is the unit disk centered at the origin. Thus, we can write $p = \varphi(q)$ where $q = (1, 1)/\sqrt{3}$. Using the definition of the tangent space, we get

$$T_q\mathbb{S}^2 = T_q\varphi_N = \text{Im} \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix} \right) = \text{span} \left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right).$$

We are now ready to define what it means for a surface to be orientable through the next two definitions.

Definition (Unit Normal). We say that $n \in \mathbb{R}^3$ is a *unit normal* to a surface S at $p \in S$ if $\|n\| = 1$ and $n \perp T_pS$.

Notice that there are exactly two unit normal at each point p (the subspace $T_pS^\perp$ of $\mathbb{R}^3$ is one-dimensional).

Definition (Orientation of a Surface, Gauss Map). An *orientation* on S is a continuous function $N : S \rightarrow \mathbb{S}^2$ such that $N(p)$ is a unit normal to S at p for all $p \in S$. We say that S is *orientable* if there exists an orientation on S . The map N is sometimes called the *Gauss map*.

Notice that any connected orientable surface has precisely two orientations. We can think of the orientation of a surface as a set of hairs on the surface either pointing inside or outside that surface. With this way of thinking about it, it may seem like every surface is orientable. It turns out that this is not the case. Let's look at an example.

Define the path $\gamma : \mathbb{R} \rightarrow \mathbb{R}^3$ by $\gamma(t) = 2(\cos t, \sin t, 0)$ and consider the *Möbius strip* parametrized by $\varphi : (-1, 1) \times \mathbb{R} \rightarrow \mathbb{R}^3$ which is defined by

$$\varphi(s, t) = \gamma(t) + s \cos \left(\frac{t}{2} \right) N(t) + s \sin \left(\frac{t}{2} \right) B(t) = \begin{pmatrix} (2 - s \cos(\frac{t}{2})) \cos t \\ (2 - s \cos(\frac{t}{2})) \sin t \\ -s \sin(\frac{t}{2}) \end{pmatrix}.$$

Next, computing the Jacobian at $(0, t)$ gives us

$$J_\varphi(0, t) = \begin{pmatrix} -\cos t \cos(\frac{t}{2}) & -2 \sin t \\ -\sin t \cos(\frac{t}{2}) & 2 \cos t \\ \sin(\frac{t}{2}) & 0 \end{pmatrix}.$$

From this, we can define a unit normal by taking the cross product of the two columns:

$$N(t) = \left(-2 \cos t \sin \left(\frac{t}{2} \right), -2 \sin t \sin \left(\frac{t}{2} \right), -2 \cos \left(\frac{t}{2} \right) \right).$$

Finally, notice that $N(t) \rightarrow (0, 0, -2)$ as $t \rightarrow 0$ but $N(t) \rightarrow (0, 0, 2)$ as $t \rightarrow 2\pi$. Hence, we get an orientation which is continuous except at 0. Thus, the Möbius strip is not orientable.

This example should not be discouraging because it turns out that most surfaces we are going to consider are indeed orientable. This is the content of the next theorem.

Proposition 3.2.2. If $V \subset \mathbb{R}^3$ is open and $F : V \rightarrow \mathbb{R}$ is C^1 and $a \in F(V)$ is a regular value of F , then $S = F^{-1}(a)$ is orientable.

Proof. Simply define the function $N : S \rightarrow \mathbb{R}^3$ by

$$N(p) = \frac{\nabla F(p)}{\|\nabla F(p)\|}.$$

Since the gradient is perpendicular to the level sets, then this is a unit normal for all p . Moreover, since F is C^1 , then N is C^0 . Therefore, N is an orientation for S , and so S is orientable. ■

3.3 Functions between Surfaces

The next step towards our understanding of surfaces is to develop the idea of functions between surfaces. This situation already appeared for curves with the curvature for example, but it gets more complicated with surfaces and so it is necessary to treat the situation rigorously.

Definition (Differentiability of a function defined on a surface). Let $S \subseteq \mathbb{R}^3$ be a regular C^k surface. The function $f : S \rightarrow \mathbb{R}^n$ is C^k if for all $p \in S$, there is a local parametrization $\varphi : U \rightarrow S$ near p such that $f \circ \varphi : U \rightarrow \mathbb{R}^n$ is C^k . The expression $f \circ \varphi$ is called f in *local coordinates*.

It follows that we can now consider the differentiability of functions between two surfaces. The following lemma lets us determine when such functions are differentiable.

Lemma 3.3.1. Let $S, \tilde{S} \subseteq \mathbb{R}^3$ be regular C^k surfaces. A function $f : \tilde{S} \rightarrow S$ is C^k if and only if for all $\tilde{p} \in \tilde{S}$, there are local parametrizations $\tilde{\varphi} : \tilde{U} \rightarrow \tilde{S}$ near $\tilde{p}$ and $\varphi : U \rightarrow S$ near $f(\tilde{p})$ such that $\varphi^{-1} \circ f \circ \tilde{\varphi} : \tilde{U} \rightarrow U$ is C^k .

Naturally, since we are dealing with differentiable functions, we would want to talk about their derivative. Fortunately, this is a well-defined notion.

Proposition 3.3.2. Let $f : \tilde{S} \rightarrow S$ be a C^1 function between regular C^1 surfaces, then for all $p \in \tilde{S}$, the linear map

$$d(f \circ \varphi)_x \circ (d\varphi_x)^{-1} : T_p \tilde{S} \rightarrow T_{f(p)} S$$

where $\varphi : \tilde{U} \rightarrow \tilde{S}$ and $x = \varphi^{-1}(p)$, is independent of the choice of surface patch φ .

The proof is just an application of the Inverse Function Theorem and of the Chain Rule. The consequence of this theorem is that it lets us define the derivative of f .

Definition. Let $f : \tilde{S} \rightarrow S$ be a C^1 function between regular C^1 surfaces, let $p \in \tilde{S}$, then we define the *derivative* of f at p as the linear map

$$df_p = d(f \circ \tilde{\varphi})_x \circ (d\tilde{\varphi}_x)^{-1} : T_p \tilde{S} \rightarrow T_{f(p)} S$$

where $\tilde{\varphi} : \tilde{U} \rightarrow \tilde{S}$ and $x = \tilde{\varphi}^{-1}(p)$.

To illustrate what we did so far, let's focus on an example. Take the unit sphere $\mathbb{S}^2 \subseteq \mathbb{R}^3$ and define the stereographic projection $\pi_N : \mathbb{R}^2 \rightarrow \mathbb{S}^2$ based at the north pole $N = (0, 0, 1)$ by

$$\pi_N(x, y) = \left(\frac{4x}{x^2 + y^2 + 4}, \frac{4y}{x^2 + y^2 + 4}, 1 - \frac{8}{x^2 + y^2 + 4} \right).$$

This map is smooth. Next, let $c \in \mathbb{R}^2$ and let $M_c : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the translation-by- c map (which is also smooth). Finally, define the function $f_c : \mathbb{S}^2 \rightarrow \mathbb{S}^2$ by

$$f_c(p) = \begin{cases} N & p = N, \\ \pi_N \circ M_c \circ \pi_N^{-1}(p) & p \neq N. \end{cases}$$

Let's show that f_c is smooth. Clearly, for all points $p \in \mathbb{S}^2 \setminus \{(0, 0, 1)\}$, it suffices to look at the function $f \circ \pi_N = \pi_N \circ M_c$ which is clearly smooth. Hence, it suffices to prove it for the north pole N . If we find the stereographic projection about the south pole $S = (0, 0, -1)$, we get the surface patch

$$\pi_S(x, y) = \left(\frac{4x}{x^2 + y^2 + 4}, \frac{4y}{x^2 + y^2 + 4}, \frac{8}{x^2 + y^2 + 4} - 1 \right)$$

which is again smooth. Since

$$(\pi_N^{-1} \circ \pi_S)(x, y) = \left(\frac{4x}{x^2 + y^2}, \frac{4y}{x^2 + y^2} \right),$$

for $(x, y) \neq (0, 0)$, then

$$(f_c \circ \pi_S)(x, y) = \left(\frac{8x}{h(x, y)}, \frac{8y}{h(x, y)}, 1 - \frac{8(x^2 + y^2)}{h(x, y)} \right)$$

where $h(x, y) = 16 + 8(xc_x + yc_y) + (4 + c_x^2 + c_y^2)(x^2 + y^2)$ and $c = (c_x, c_y)$. This expression is only valid for $(x, y) \neq (0, 0)$ because f_c is defined differently for $(x, y) = (0, 0)$. However, notice that if we plug-in $x = y = 0$ in the expression above, we also get N as the output. Therefore, the equation

$$(f_c \circ \pi_S)(x, y) = \left(\frac{8x}{h(x, y)}, \frac{8y}{h(x, y)}, 1 - \frac{8(x^2 + y^2)}{h(x, y)} \right)$$

is valid for all $(x, y) \in \mathbb{R}^2$. Next, this expression is smooth at $(0, 0)$ because $h(0, 0) = 16 \neq 0$ and all the quantities involved are smooth. Therefore, f_c is smooth at $N =$

$\pi_S(0,0)$. It follows that f_c is smooth on its domain.

We will end this section by defining the type of map between surfaces that will be of interest. In the same way that we consider morphisms, i.e. maps that preserve some operation, in algebra, we will now consider maps that locally preserve the length of the vectors in the tangent space.

Definition (Local Isometry, Isometry). Let $f : \tilde{S} \rightarrow S$ be a C^1 function between two regular C^1 surfaces. The function f is a *local isometry* if $\|v\|^2 = \|df_p(v)\|^2$ for all $v \in T_p S$ and $p \in S$. If in addition f is a bijection, then it is an *isometry*. If there is an isometry between two surfaces, then the surfaces are *isometric*.

Notice that if f is a restriction of a rigid motion of $\mathbb{R}^3$, then f is necessarily an isometry (which should make sense intuitively).

3.4 The First Fundamental Form

In this section, we will see how to compute lengths and areas of curves and regions on surfaces. To do this, we need to define the First Fundamental Form, an object which is central to the study of surfaces.

Definition (First Fundamental Form). Let $S \subseteq \mathbb{R}^3$ be a regular C^1 surface. The *First Fundamental Form* assigns to each $p \in S$ the quadratic form $\|\cdot\|^2 : T_p S \rightarrow \mathbb{R}$, the norm is the usual norm in $\mathbb{R}^3$.

However, this definition is global. It will be better for us to work in local coordinates. That is why we need the following definition.

Definition. Let $\varphi : U \rightarrow S$ be a surface patch of a regular C^1 surface. For each $q \in U$, define the functions

$$I_q : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad u \mapsto \|d\varphi_q(u)\|^2$$

and

$$\langle \cdot, \cdot \rangle_q : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (u_1, u_2) \mapsto \frac{1}{4}(I_q(u_1 + u_2) - I_q(u_1 - u_2)).$$

Next, consider the same set-up as in the previous definition, denote by u and v the two variables of φ , and take the vector $(x, y) \in \mathbb{R}^2$, then

$$I_q(x, y) = \|d\varphi_q(x, y)\|^2 = \left\| \begin{pmatrix} \frac{\partial}{\partial u} \varphi(q) & \frac{\partial}{\partial v} \varphi(q) \\ \vdots & \vdots \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \right\|^2 = \|x \partial_u \varphi(q) + y \partial_v \varphi(q)\|^2.$$

Now, if we expand the norm squared, we get

$$I_q(x, y) = \|\partial_u \varphi(q)\|^2 x^2 + 2 \partial_u \varphi(q) \cdot \partial_v \varphi(q) xy + \|\partial_v \varphi(q)\|^2 y^2.$$

Finally, if we write $E(q) = \|\partial_u \varphi(q)\|^2$, $F(q) = \partial_u \varphi(q) \cdot \partial_v \varphi(q)$ and $G(q) = \|\partial_v \varphi(q)\|^2$, and denote by du the function that outputs the first coordinate of a vector, and by dv the function the outputs the second coordinate of a vector, we get that

$$I_q = E(q)du^2 + 2F(q)dudv + G(q)dv^2.$$

This explains why the First Fundamental Form is commonly described as

$$ds^2 = Edu^2 + 2Fdudv + Gdv^2.$$

TODO

- (a) Explain the link between local isometries and the first fundamental form: $I_p(v) = I_{f(p)}(df_p(v))$.
- (b) Show examples of local isometries using first fundamental form
- (c) Compute lengths using first fundamental form
- (d) Compute areas using first fundamental form

TODO

Now, let's compute length of "curves" in the surface. To do this, we can find a local parametrization, and compute the usual length in $\mathbb{R}^2$ of the projection in local coordinates of the curves. However, if we have an other local parametrization, the projection of the same curve may not have the same length in $\mathbb{R}^2$ under this second parametrization. To fix this problem, we will use the first fundamental form.

Example 1: Cylinders. Take $S_1 = \{x^2 + y^2 = 1\}$ parametrized by $\varphi_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $\varphi(s, t) = (\cos s, \sin s, t)$. Let's compute the first fundamental form for $q = (s, t)$.

$$J_{\varphi_1}(q) = \begin{pmatrix} -\sin s & 0 \\ \cos s & 0 \\ 0 & 1 \end{pmatrix}.$$

Hence, $E(q) = 1$, $F(q) = 0$ and $G(q) = 1$, and so I_1 looks like $ds^2 + dt^2$ which is the Euclidean metric. Thus, to compute the length of a curve on the cylinder, it suffices to project the curve on the s, t plane and compute the length there. Thus, to find the shortest path between two points on the cylinder, we need to take the image of the straightline between the two corresponding points in the s, t plane. Let's generalize this. Let $C \subseteq \mathbb{R}^2$ be a regular closed curve with a unit speed parametrization $\gamma : \mathbb{R} \rightarrow \mathbb{R}^2$. Define $S_2 = \{(x, y, z) : (x, y) \in C\}$ and parametrize by $\varphi_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $\varphi_2(s, t) = (\gamma(s), t)$. The same computation as above shows that $I_2 = ds^2 + dt^2$. So the same conclusions follow. Since S_1 and S_2 have the same first fundamental form, then they are locally isometric.

Example 2: S_1 us parametrized by $\psi : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^3$ where

$$\psi(u, v) = \left(\frac{u}{\sqrt{u^2 + v^2}}, \frac{v}{\sqrt{u^2 + v^2}}, \frac{1}{2} \ln(u^2 + v^2) \right).$$

The image of ψ is a cylinder of radius 1 centered at the z -axis in $\mathbb{R}^3$. The image of the unit circle is the unit circle. The image of the circles of radius $r < 1$ are the unit circle below the xy -plane. Similar for the circles of bigger radius. Here, if we take $q = (u, v)$, we get

$$J_{\psi}(q)^T J_{\psi}(q) = \frac{1}{u^2 + v^2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Hence, the first fundamental form is $I_q = \frac{du^2+dv^2}{u^2+v^2}$. To compute the magnitude of a point, we need to compute the square root of the fundamental form at that point.

Let's compute length/area in local coordinates. Let $C = \{u^2+v^2 = r^2\}$ parametrized by $(r \cos t, r \sin t)$.

$$\ell(C) = \int_C C ds_\psi = \int_0^{2\pi} \sqrt{\frac{du^2+dv^2}{u^2+v^2}} = \int_0^{2\pi} \sqrt{\frac{(-r \sin t)^2 + (r \cos t)^2}{(r \cos t)^2 + (r \sin t)^2}} dt = 2\pi$$

which makes sense following our geometric intuition.

TODOTODOTODOTODOTODO

3.5 Gaussian and Mean Curvature

Definition (Shape Operator). Let $S \subset \mathbb{R}^3$ be an orientable regular C^2 surface with orientation $N : S \rightarrow \mathbb{S}^2$. For $p \in S$, define the *shape operator* at p by

$$-dN_p : T_p S \rightarrow T_{N(p)} \mathbb{S}^2 = T_p S.$$

Definition (Gaussian and Mean Curvature). Let $S \subset \mathbb{R}^3$ be an orientable regular C^2 surface with orientation $N : S \rightarrow \mathbb{S}^2$. For $p \in S$, define the *Gaussian curvature* and the *Mean curvature* by

$$K(p) = \det(-dN_p), \quad H(p) = \frac{1}{2} \text{Tr}(-dN_p).$$

Proposition 3.5.1. The Gaussian and Mean curvatures are preserved under rigid motions.

Proposition 3.5.2. The shape operator is self-adjoint.

Corollary. Let $S \subset \mathbb{R}^3$ be a regular C^2 surface with orientation $N : S \rightarrow \mathbb{S}^2$. For all $p \in S$, there is an orthonormal basis $\{\vec{v}_1, \vec{v}_2\}$ of $T_p S$ such that

$$-dN_p = \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix}$$

where we assume that $k_1 \leq k_2$. The orthonormal basis induces the *principal directions*. The eigenvalues k_1 and k_2 are called the principal values.

Corollary.

$$K(p) = k_1 k_2, \quad H(p) = \frac{k_1 + k_2}{2}.$$

Definition. (a) p is elliptic if $K(p) > 0$. (sphere, $z = x^2 + y^2$)

(b) p is hyperbolic if $K(p) < 0$. (curve with saddle point)

(c) p is parabolic if $K(p) = 0$ but $H(p) \neq 0$. ($y = z^2$ in $\mathbb{R}^3$)

(d) p is planar if $K(p) = H(p) = 0$. (a plane)

(e) p is umbilic if $k_1 = k_2$. (sphere)

3.6 The Second Fundamental Form

Definition. Let $S \subset \mathbb{R}^3$ be a regular C^2 surface with orientation $N : S \rightarrow \mathbb{S}^2$ and let $p \in S$. The *Second Fundamental Form* of S at p is the function $\Pi_p : T_p S \rightarrow \mathbb{R}$ defined by

$$\Pi_p(v) = -dN_p(v) \cdot v.$$

Proposition 3.6.1. The shape operator $-dN_p : T_p S \rightarrow T_p S$ (wrt to $\{\partial_u \varphi(p), \partial_v \varphi(p)\}$) is represented by the matrix

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix}^{-1} \begin{pmatrix} e & f \\ f & g \end{pmatrix}.$$

Corollary (Curvatures in local coordinates).

$$K \circ \varphi = \frac{eg - f^2}{EG - F^2}, \quad 2H \circ \varphi = \frac{eG - 2fF + gE}{EG - F^2}.$$

TODOTODOTODO

[I skipped many classes. The following is from March the 12th]

Goal: Let $S \subset \mathbb{R}^3$ be a regular C^2 surface and $K : S \rightarrow \mathbb{R}$ its Guassian curvature. $K \equiv 0$ if and only if S is locally parametrized by a local isometry.

Exercise: the double cone has $K \equiv 0$. **TODOTODOTODO**

Some Linear Algebra. Let V be a finite dimensional vector space over $\mathbb{R}$, $L : V \rightarrow V$ a linear transformation, and $\langle \cdot, \cdot \rangle$ an innerproduct on V . Definition: L is self-adjoint is $\langle Lv, w \rangle = \langle v, Lw \rangle$ for all $v, w \in V$. TFAE:

- (a) L is self-adjoint
- (b) L is represented by a diagonal matrix with respect to some orthogonal basis of V
- (c) L is represented by a symmetric matrix with respect to some orthogonal basis of V
- (d) The bilinear form $\langle \langle \cdot, \cdot \rangle \rangle_L$ on V is symmetric ($\langle \langle v, w \rangle \rangle_L = \langle \langle Lv, w \rangle \rangle$).

Lemma 3.6.2 (Shape operator is self-adjoint). Let $S \subset \mathbb{R}^3$ be a regular C^2 surface with orientation $N : S \rightarrow \mathbb{S}^2$. If $p \in S$, then $-dN_p : T_p S \rightarrow T_p S$ is self-adjoint: $-dN_p(v) \cdot w = v \cdot (-dN_p(w))$ for all $v, w \in T_p S$.

Proof. **TODOTODOTODO**[reread the proof] **TODOTODOTODO**If $p \in S$ is a critical point and $-dN_p$ is represented by Hessian, we are done since Hessian is symmetric. Otherwise, choose a local parametrization $\varphi : U \rightarrow S$ near $p \in S$. Notice that it is enough to prove the self-adjoint condition for a basis of $T_p S$. We need to show that $-dN_p(\partial_u \varphi) \cdot \partial_v \varphi = \partial_u \varphi \cdot (-dN_p(\partial_v \varphi))$. Observe that $N \circ \varphi \perp \partial_u \varphi$ on U . Hence, $N \circ \varphi \cdot \partial_u \varphi \equiv 0$. Taking the derivative with respect to v on both sides:

$$\begin{aligned} \partial_v(N \circ \varphi) \cdot \partial_u \varphi + N \circ \varphi \cdot \partial_{uv}^2 \varphi &\equiv 0 \\ \implies \partial_v(N \circ \varphi) \cdot \partial_u \varphi &\equiv -N \circ \varphi \cdot \partial_{uv}^2 \varphi \end{aligned}$$

By the same argument $\partial_u(N \circ \varphi) \cdot \partial_v \varphi = -N \circ \varphi \cdot \partial_{uv}^2 \varphi$. Thus,

$$-\partial_v(N \circ \varphi) \cdot \partial_v \varphi = -\partial_u(N \circ \varphi) \cdot \partial_v \varphi.$$

Finally, $\partial_v(N \circ \varphi) = dN_p(\partial_v \varphi)$ and $\partial_u(N \circ \varphi) = dN_p(\partial_u \varphi)$ so we get

$$-dN_p(\partial_u \varphi) \cdot \partial_v \varphi = \partial_u \varphi \cdot (-dN_p(\partial_v \varphi)).$$

TODO ■

Corollary. The shape operator is diagonalizable: $-dN_p = \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix}$ where we assume that $k_1 \leq k_2$.

Definition. The second fundamental form of S at p is the function $\text{II}_p : T_p S \rightarrow \mathbb{R}$ defined by $\text{II}_p(v) = -dN_p(v) \cdot v$. In local coordinates, using the local parametrization $\varphi : U \rightarrow S$, it is represented by a symmetric matrix $\begin{pmatrix} e & f \\ f & g \end{pmatrix}$. We denote II_p by $edu^2 + 2fdudv + gdv^2$. Here,

$$\begin{aligned} e &= -dN_p(\partial_u \varphi) \cdot \partial_u \varphi = N \circ \varphi \cdot \partial_{uu}^2 \varphi \\ f &= -dN_p(\partial_v \varphi) \cdot \partial_u \varphi = N \circ \varphi \cdot \partial_{uv}^2 \varphi \\ g &= -dN_p(\partial_v \varphi) \cdot \partial_v \varphi = N \circ \varphi \cdot \partial_{vv}^2 \varphi \end{aligned}$$

The last term in every line is more convenient to compute the coefficients of the second fundamental form because we only need to compute the partials of φ , not N .

Proposition 3.6.3 (Shape operator in local coordinates). The shape operator $-dN_p : T_p S \rightarrow T_p S$ is represented by the 2x2 matrix

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix}^{-1} \begin{pmatrix} e & f \\ f & g \end{pmatrix} = \frac{1}{EG - F^2} \begin{pmatrix} eG - fF & fG - gF \\ fE - eF & gE - fF \end{pmatrix}$$

with respect to the basis $\{\partial_u \varphi, \partial_v \varphi\}$ where everything is evaluated at q .

Proof. Represent the shape operator by $\begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{pmatrix}$ and notice that

$$\begin{aligned} e(q) &= -dN_p(\partial_u \varphi(q)) \cdot \partial_u \varphi(q) \\ &= (a_{11} \partial_u \varphi + a_{21} \partial_v \varphi) \cdot \partial_u \varphi|_q \\ &= a_{11} E(q) + a_{21} F(q) \end{aligned}$$

Similarly, $f(q) = a_{11} F(q) + a_{21} G(q)$. It follows that

$$\begin{pmatrix} e \\ f \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}.$$

Similarly, $f(q) = a_{12} E(q) + a_{22} F(q)$ and $g(q) = a_{12} F(q) + a_{22} G(q)$ implies that

$$\begin{pmatrix} f \\ g \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}.$$

Therefore,

$$\begin{pmatrix} e & f \\ f & g \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}.$$

The desired formula follows by taking the inverse of the first fundamental form. ■

Corollary (Curvatures in local coordinates).

$$K \circ \varphi = \frac{eg - f^2}{EG - F^2}, \quad 2H \circ \varphi = \frac{eG - 2fF + gE}{EG - F^2}.$$

Positive Gaussian curvature at a point can be seen as the normal preserving the orientation of a circle around the point, and negative Gaussian curvature at a point can be seen as the normal reversing the orientation of a circle around the point.

Definition. The *normal curvature* of S at p along the unit vector $v \in T_p S$ is defined as $\Pi_p(v)$.

Proposition 3.6.4 (Characterization of normal curvature). Let $S \subset \mathbb{R}^3$ be a regular C^2 surface with orientation $N : S \rightarrow \mathbb{S}^2$, $p \in S$ and v a unit vector in the tangent space $T_p S$. Take $\gamma : (-\epsilon, \epsilon) \rightarrow S$ be a regular C^2 path with $\gamma(0) = p$ and $\dot{\gamma}(0) = v$. Then the normal curvature along v can be computed as $N(p) \cdot \ddot{\gamma}(0)$, i.e., $\Pi_p(v) = N(p) \cdot \ddot{\gamma}(0)$. Moreover, $|\Pi_p(v)| \leq \kappa_\gamma(0)$ with equality achieved by normal section along v (normal section is the intersection between surface and the plane spanned by normal vector and v).